

CPS Flood Decision Support

Montpelier, VT

271 buildings

PLAN / FORECAST

Damage Probability — 100-year Flood

22.5 ft crest - 271 pre-code masonry buildings

No damage

Cosmetic

Structural

PLAN / FORECAST

UPDATE MODEL

FLOOD SCENARIO

2023

2024

100-yr

50-yr

10-yr

Custom

NWS Live

20

NO DAMAGE

121

COSMETIC

5

STR

Under a 100-year flood (22.5 ft), buildings on Main Street have a 88% probability of damage exceeding \$15,000. The proposed retention basin reduces that to 56%.

WHAT-IF: EXPECTED LOSS

No action

\$7.3M

Retention basin

\$5.4M

Bldg. hardening

\$5.7M

Combined

\$4.2M

CPS Flood Decision Support

Montpelier, VT

271 buildings

PLAN / FORECAST

With Retention Basin — 100-year Flood

22.5 ft crest - upstream retention basin applied

No damage

Cosmetic

Structural

PLAN / FORECAST

UPDATE MODEL

FLOOD SCENARIO

2023

2024

100-yr

50-yr

10-yr

Custom

NWS Live

30

NO DAMAGE

166

COSMETIC

3

STR

The upstream retention basin reduces expected loss by ↓ 26% across 271 buildings, shifting 55 buildings from structural to cosmetic or no damage.

WHAT-IF: EXPECTED LOSS

No action

\$7.3M

Retention basin

\$5.4M

Bldg. hardening

\$5.7M

Combined

\$4.2M

CPS Flood Decision Support

Montpelier, VT

271 buildings

PLAN / FORECAST

Damage Probability — NWS Forecast

NWS forecast - 48-hour lead - 17.2 ft crest - 90% credible intervals

No damage

Cosmetic

Structural

PLAN / FORECAST

UPDATE MODEL

FLOOD SCENARIO

2023

2024

100-yr

50-yr

10-yr

Custom

NWS Live

159

NO DAMAGE

112

COSMETIC

5

STR

NWS 48-hour forecast predicts 17.2 ft crest. 0 buildings at structural risk; 112 at cosmetic risk. Pre-storm staging priorities are shown below.

PRE-STORM STAGING PRIORITY

1

Main St / Bridge St

Deploy flood barrier

2

State St corridor

Sandbag basement entries

3

Langdon St (N. Branch)

Monitor preparation

4

Elm St storm drains

Clear debris

WHAT-IF: EXPECTED LOSS

No action

\$1.3M

Retention basin

\$1.0M

Bldg. hardening

\$1.1M

Combined

\$0.8M